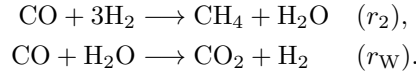


BARNOS PART

Governing balances, axial profiles, and operating-condition sensitivity for CO methanation.

Model basis

The bed is treated as a steady, one-dimensional plug-flow reactor. The reduced network retains CO methanation and water-gas shift (WGS):



At each position, the reduced M4 kinetics give net rates r_2 and r_W from local temperature and partial pressures $p_i = y_i P$. Rates are in $\text{mol kg}_{\text{cat}}^{-1} \text{s}^{-1}$, flows F_i in mol s^{-1} , and a is catalyst activity. Integration uses catalyst mass W . For uniform loading ρ_B in a tube of cross-section $A_c = \pi D_t^2/4$,

$$W = \rho_B A_c z, \quad z = \frac{W}{\rho_B A_c}, \quad W_{\text{tot}} = \rho_B A_c L.$$

A steady mole balance over a catalyst slice dW gives $dF_i/dW = a \sum_j \nu_{ij} r_j$, where a is catalyst activity. Written for the modeled species:

$$\begin{aligned}\frac{dF_{\text{CO}}}{dW} &= -a(r_2 + r_W), & \frac{dF_{\text{H}_2}}{dW} &= a(-3r_2 + r_W), \\ \frac{dF_{\text{CH}_4}}{dW} &= ar_2, & \frac{dF_{\text{H}_2\text{O}}}{dW} &= a(r_2 - r_W), \\ \frac{dF_{\text{CO}_2}}{dW} &= ar_W, & \frac{dF_{\text{N}_2}}{dW} &= 0.\end{aligned}$$

Temperature from the energy balance

For a differential slice, reaction heat raises the gas sensible enthalpy; heat transfer to the wall removes heat when the gas is hotter than the wall. Thus

$$\sum_i F_i C_{p,i} \frac{dT}{dW} = a [r_2(-\Delta H_2) + r_W(-\Delta H_W)] - \frac{U a_w}{\rho_B} (T - T_w).$$

Dividing by the gas heat-capacity flow gives the temperature equation integrated by the solver:

$$\frac{dT}{dW} = \frac{a [r_2(-\Delta H_2) + r_W(-\Delta H_W)] - \frac{U a_w}{\rho_B} (T - T_w)}{\sum_i F_i C_{p,i}}$$

Here $a_w = 4/D_t$ is the cylindrical heat-transfer area per bed volume. The formulation uses exothermic heats $-\Delta H_2 = 206$ and $-\Delta H_W = 41 \text{ kJ mol}^{-1}$. The current implementation uses constant heat capacities and these constant reaction enthalpies.

Pressure from the Ergun equation

The Ergun pressure gradient combines viscous resistance (first term) and inertial resistance (second term):

$$\frac{dP}{dz} = - \left[\frac{150\mu(1-\varepsilon_b)^2 u_s}{D_p^2 \varepsilon_b^3} + \frac{1.75\rho_g(1-\varepsilon_b)u_s^2}{D_p \varepsilon_b^3} \right].$$

The reacting-gas properties are recalculated locally:

$$u_s = \frac{F_{\text{tot}} RT}{P A_c}, \quad \rho_g = \frac{P \bar{M}}{RT}, \quad \bar{M} = \sum_i y_i M_i, \quad y_i = \frac{F_i}{F_{\text{tot}}}.$$

Since $dz/dW = 1/(\rho_B A_c)$, the pressure equation in the solver's catalyst-mass coordinate is

$$\frac{dP}{dW} = - \frac{1}{\rho_B A_c} \left[\frac{150\mu(1-\varepsilon_b)^2 u_s}{D_p^2 \varepsilon_b^3} + \frac{1.75\rho_g(1-\varepsilon_b)u_s^2}{D_p \varepsilon_b^3} \right]$$

Pressure is integrated in Pa; the kinetic partial pressures are evaluated in bar. The pressure loss is coupled to reaction because the local flow, composition, temperature, and density change down the bed.

Integrated solution and plotting

The numerical problem is the coupled initial-value system

$$\frac{dy}{dW} = \mathbf{f}(\mathbf{y}), \quad \mathbf{y}(W) = \begin{bmatrix} \mathbf{F}(W)^\top & T(W) & P(W) \end{bmatrix}^\top, \quad \mathbf{y}(0) = \begin{bmatrix} \mathbf{F}_0^\top & T_0 & P_0 \end{bmatrix}^\top, \quad 0 \leq W \leq W_{\text{tot}}.$$

The code integrates this system with the stiff Radau method and samples the solution at 401 evenly spaced catalyst-mass locations. The plotted coordinates are $z_j = W_j/(\rho_B A_c)$, $T_j = T(W_j)$, and $P_{j,\text{bar}} = P(W_j)/10^5$. CO conversion is calculated as

$$X_{\text{CO}}(W) = 1 - \frac{F_{\text{CO}}(W)}{F_{\text{CO},0}}.$$

Differentiating this definition and substituting the CO balance gives

$$\frac{dX_{\text{CO}}}{dW} = \frac{a(r_2 + r_W)}{F_{\text{CO},0}}.$$

Therefore, the figures show samples of the integrated profiles, not a separate closed-form solution for $T(z)$ or $P(z)$.

Plotted demonstration case. $D_t = 0.0254$ m (1.00 in), $L = 0.3048$ m (12.00 in), $\rho_B = 750$ kg m⁻³, $\varepsilon_b = 0.40$, $D_p = 3$ mm, $\mu = 2.5 \times 10^{-5}$ Pa s, $U = 10$ W m⁻² K⁻¹; $T_0 = T_w = 623.15$ K, $P_0 = 5$ bar, and feed ratio CO:H₂:N₂=1:4:25 at 0.05 mol s⁻¹. Several geometry, feed, transport, and heat-transfer inputs remain assumed; this run is illustrative, not experimental validation.

Equation basis: Subah and Somo formulation, Sections 3, 6, and 7. Numerical implementation: `src/methanation/reactor.py`; plotting: `src/methanation/reporting.py`.

CO conversion along the catalyst bed

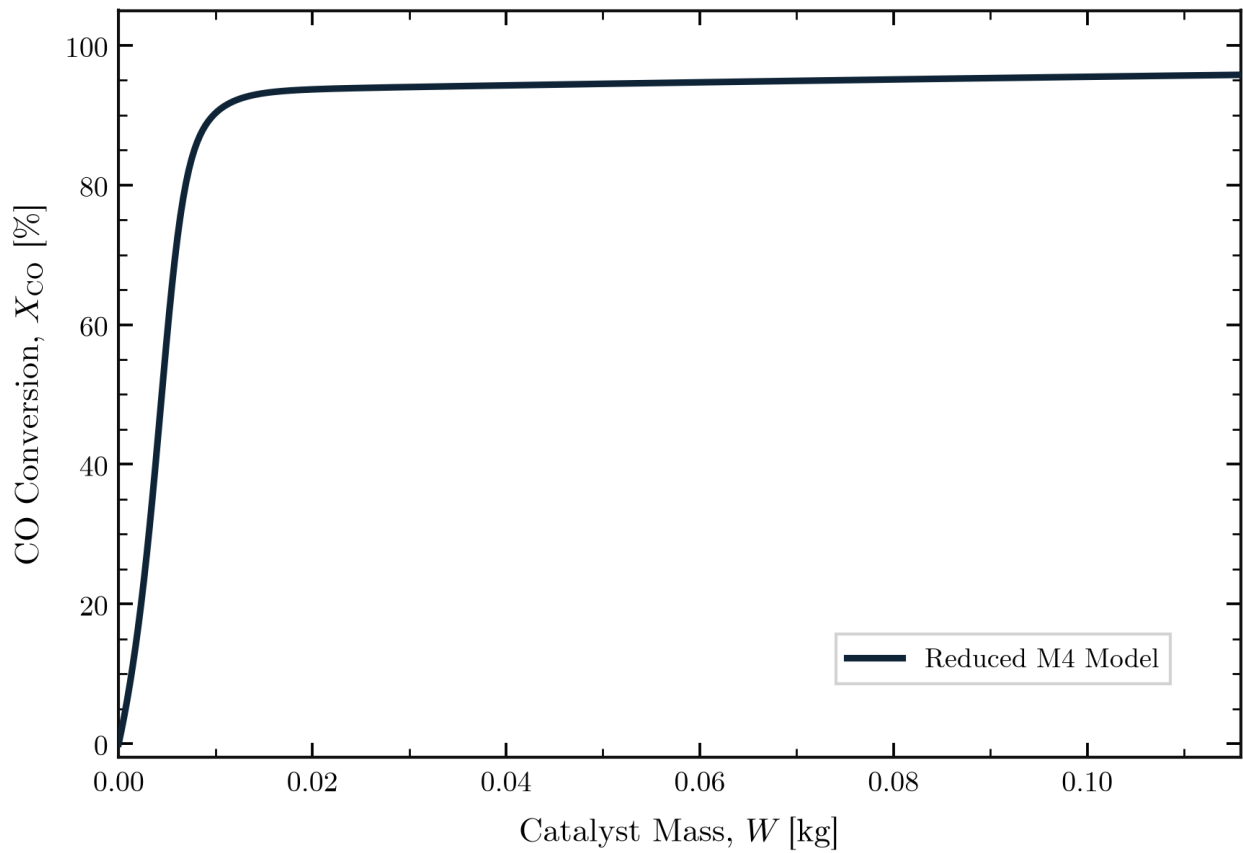


Figure 1. Reduced M4 model prediction of CO conversion versus catalyst mass for the demonstration case.

The calculated outlet conversion is 95.78% for this assumed configuration. This curve is a model demonstration, not an experimental fit or validation result.

Temperature and pressure along the bed

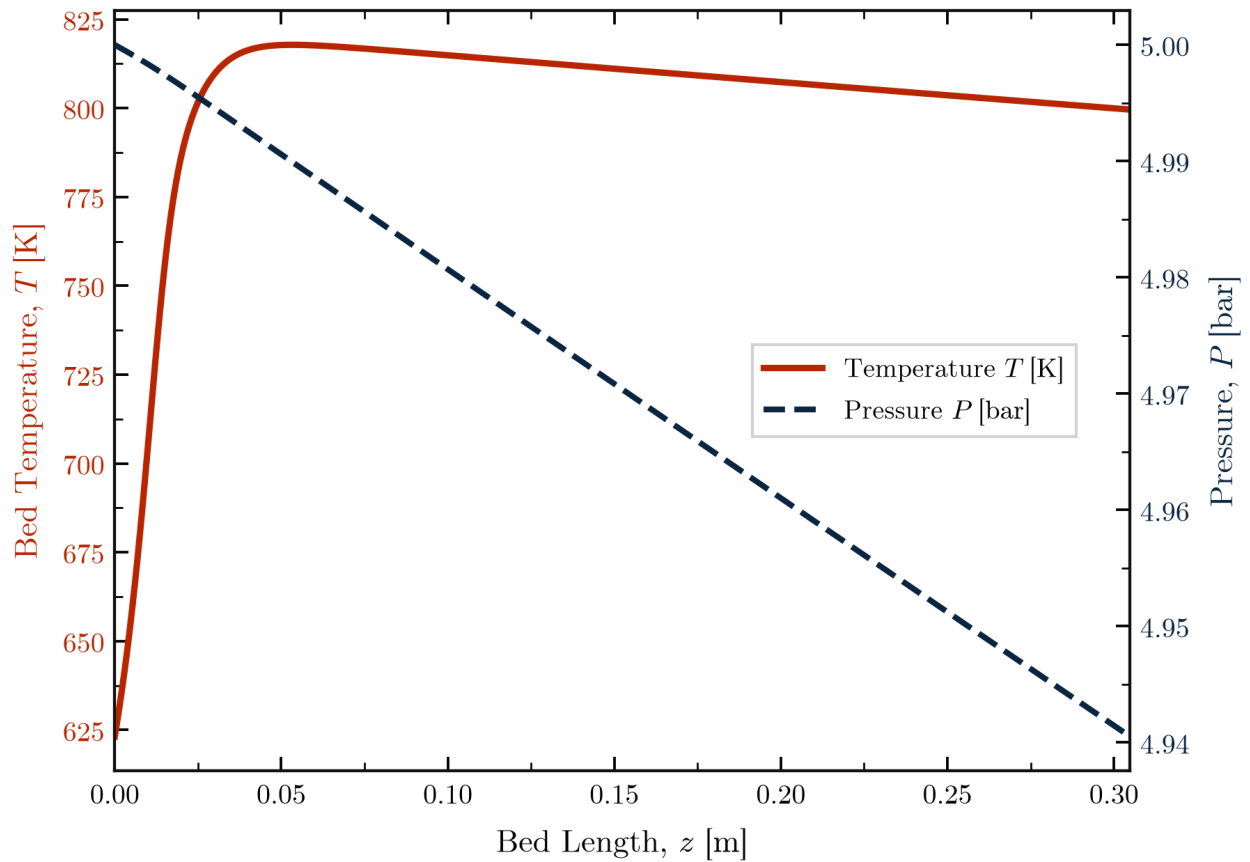


Figure 2. Integrated temperature and Ergun pressure profiles versus bed length.

The sampled temperature peaks at 817.8 K near $z = 0.0533$ m and is 799.6 K at the outlet. Pressure falls from 5.0000 to 4.9405 bar (about 5.95 kPa). The plotted pressure axis is tightly auto-scaled, so the visible line exaggerates this relative drop.

CO conversion across reactor inlet conditions

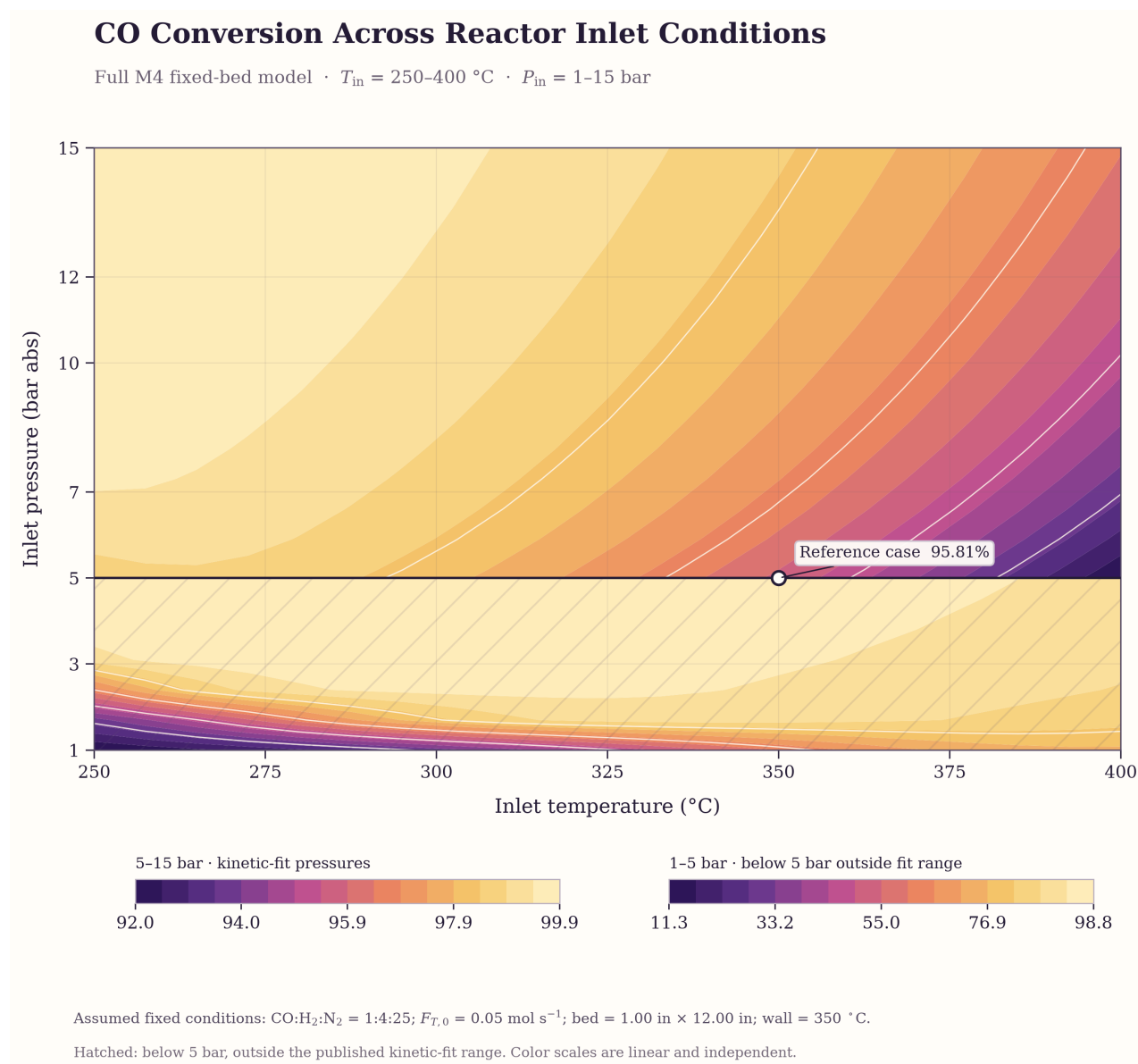


Figure 3. Full M4 sensitivity of bed-outlet CO conversion to inlet temperature and pressure.

The hatched $1 < 5$ bar region is below the pressure range tested for the Celoria et al. kinetic fit and is an extrapolation; $5\text{--}15$ bar is within the reported pressure range. The two bands use separate linear color scales, so read each colorbar independently. The sweep covers $250\text{--}400\text{ }^{\circ}\text{C}$ and varies only inlet temperature and pressure, with the other assumed base-case inputs fixed. Although inlet temperatures lie within the paper's fitted temperature range, 469 of 484 cases have sampled bed temperatures above $400\text{ }^{\circ}\text{C}$ and therefore extrapolate the kinetics locally. This is a conditional model sensitivity, not experimental validation. Kinetic source: [Celoria et al.](#)

CO conversion within the kinetic-fit pressure range

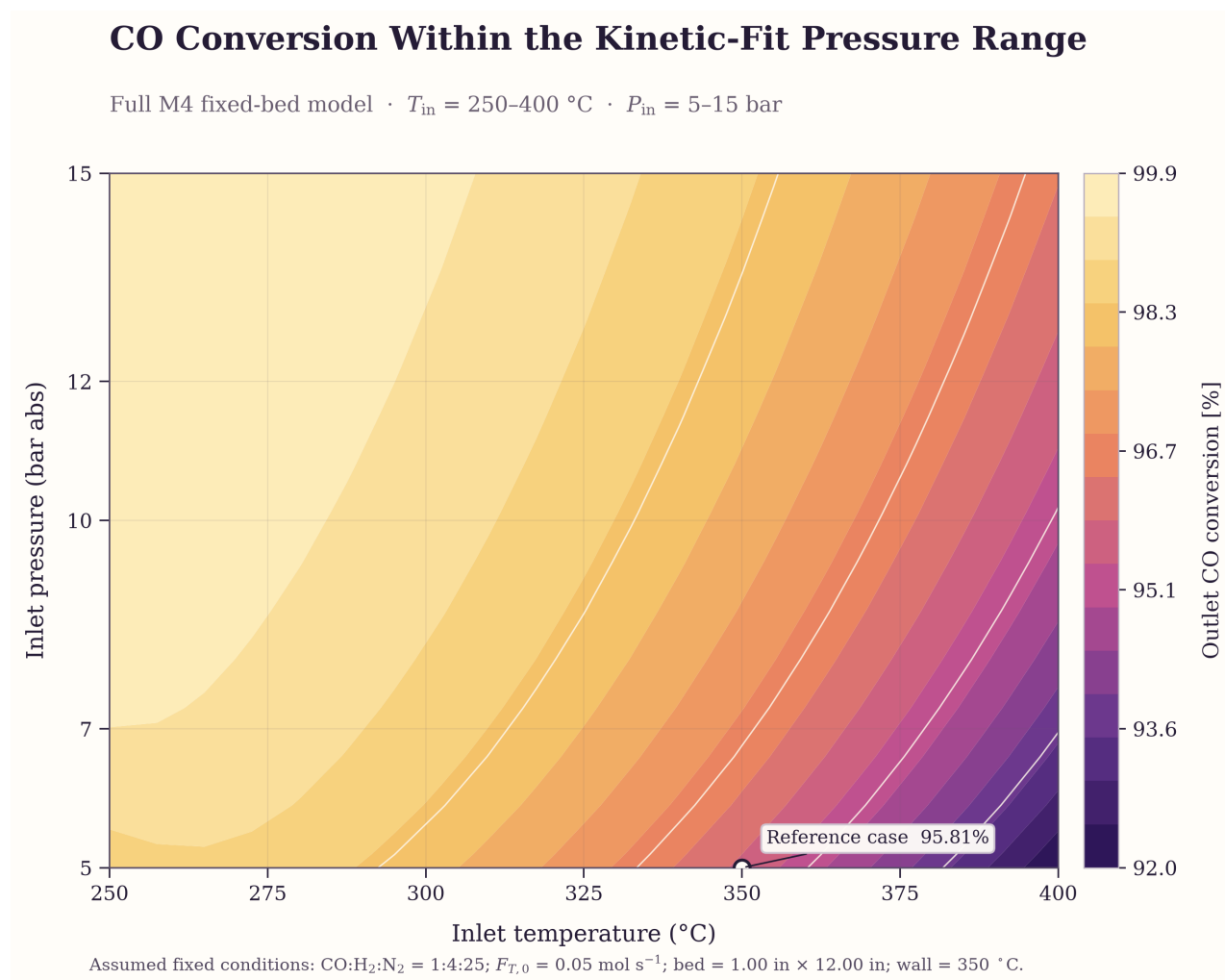


Figure 4. Full M4 sensitivity restricted to inlet pressures of 5–15 bar.

This view isolates the pressure interval reported for the Celoria et al. kinetic fit, with a single linear color scale. Inlet temperature spans 250–400 $^{\circ}\text{C}$. The kinetic fit was tested at discrete temperature and pressure conditions; values between those conditions are model interpolations. The reactor's predicted local bed temperature may exceed 400 $^{\circ}\text{C}$ even when its inlet temperature is within range, so the result remains an exploratory conditional prediction rather than experimental validation.